

Table 1: department

```
CREATE TABLE department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
INSERT INTO department VALUES  
(1, 'HR'),  
(2, 'Finance'),  
(3, 'IT'),  
(4, 'Marketing');
```

Table 2: employee

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary INT,  
    hire_date DATE,  
    city VARCHAR(50),  
    FOREIGN KEY (dept_id) REFERENCES department(dept_id)  
);
```

```
INSERT INTO employee VALUES  
(101, 'Rohit', 3, 60000, '2020-04-12', 'Delhi'),  
(102, 'Sneha', 1, 45000, '2019-09-15', 'Mumbai'),  
(103, 'Amit', 2, 55000, '2021-06-20', 'Delhi'),  
(104, 'Neha', 3, 72000, '2018-11-30', 'Pune'),  
(105, 'Vikas', 4, 38000, '2022-03-05', 'Delhi'),  
(106, 'Priya', 1, 48000, '2021-07-11', 'Chandigarh'),  
(107, 'Ankit', 3, 53000, '2023-01-19', 'Noida'),  
(108, 'Kiran', 4, 41000, '2020-05-22', 'Pune'),  
(109, 'Manish', 2, 64000, '2022-12-10', 'Delhi'),  
(110, 'Deepika', 3, 58000, '2019-02-25', 'Chandigarh');
```

Table 3: project

```
CREATE TABLE project (  
    proj_id INT PRIMARY KEY,  
    proj_name VARCHAR(50),  
    dept_id INT,  
    budget INT,  
    FOREIGN KEY (dept_id) REFERENCES department(dept_id)  
);
```

```
INSERT INTO project VALUES  
(201, 'Payroll System', 2, 800000),  
(202, 'Website Revamp', 3, 600000),  
(203, 'Recruitment Portal', 1, 400000),  
(204, 'Ad Campaign', 4, 300000),  
(205, 'Cloud Migration', 3, 900000);
```

Table 4: works_on

```
CREATE TABLE works_on (  
    emp_id INT,  
    proj_id INT,  
    hours_worked INT,  
    FOREIGN KEY (emp_id) REFERENCES employee(emp_id),  
    FOREIGN KEY (proj_id) REFERENCES project(proj_id)  
);
```

```
INSERT INTO works_on VALUES  
(101, 202, 120),  
(101, 205, 100),  
(102, 203, 150),  
(103, 201, 80),  
(104, 205, 200),  
(105, 204, 90),  
(106, 203, 60),  
(107, 202, 70),  
(108, 204, 110),  
(109, 201, 130),  
(110, 202, 85);
```

Basic SELECT & WHERE

1. Display all records from the **employee** table.
2. Show employee names and salaries only.
3. List all departments from the **department** table.
4. Display all employees who belong to the **IT department**.
5. Find employees whose salary is greater than **50,000**.
6. Display employees who live in **Delhi**.
7. Show employee names and hire dates.
8. List all projects with their budget.
9. Display projects having a budget greater than **5,00,000**.
10. Show all employees hired after **2020-01-01**.
11. Display employee name along with their **department name**.
12. List employee name and project name they are working on.
13. Show department name and project name for all projects.
14. Display employee name and hours worked on projects.
15. List all employees working on the **Website Revamp** project.
16. Count total number of employees.
17. Find the **highest salary** among employees.
18. Find the **average salary** of employees.
19. Count how many employees work in each department.
20. Show total number of projects in the company.

21. Display department name and **average salary** of employees in each department.
 22. List employees who are working on **more than one project**.
 23. Find total hours worked by each employee.
 24. Display project name and **total hours worked** on each project.
 25. List employees who are working on projects with a **budget greater than 6,00,000**.
 26. Find employees whose salary is **greater than the average salary**.
 27. Display department names that have **more than 2 employees**.
 28. List employees who are **not assigned to any project** (if any).
 29. Find the department which has the **maximum number of projects**.
 30. Display employee names who work in the **same department as Rohit**.
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1. Display all employees along with their department names.
2. List employees who work in **IT department** and have salary > 55000.
3. Show all projects handled by the **Finance** department.
4. Find total salary paid in each department.
5. Show employee names and number of projects each employee is working on.
6. Find the average salary of employees in each department.
7. Display department names having **average salary > 50000**.
8. Show employees who joined after year 2020.
9. List all employees from **Delhi** working on any project.
10. Display employees who are not assigned to any project.

11. Find employees who earn more than the **average salary** of the company.
12. Show the **highest-paid employee** in each department.
13. Display employees working on projects with **budget > 700000**.
14. Show project names where **at least 2 employees** are working.
15. Display employee(s) who have worked the **maximum hours** overall.
16. Show all employees whose salary is greater than **Rohit's salary**.
17. Find departments that have **no projects assigned**.
18. Show the **second-highest salary** in each department.
19. Display employees who work on the **same project** as 'Amit'.
20. List employees who work on **both project 202 and 205**.
21. Find total hours worked by each employee on all projects.
22. Display project names and total hours spent by employees.
23. Show departments with **more than 2 employees**.
24. Find employees who have worked **more than 100 hours** in total.
25. Show department name and total project budget handled by it.
26. Find employees who work on **projects of their own department** only.
27. List employees whose **department has a budget > 600000** in total.
28. Display department name, total salary, and total hours worked (combine data from all 3 tables).
29. Find the **top 3 highest-paid employees** along with department names.
30. Display employees grouped by city, showing **average salary per city**.

- 1. Display employees who earn more than the average salary of all employees.**
- 2. Show employees who work in departments having a budget more than ₹600000.**
- 3. Display names of employees who work on the same project(s) as 'Rohit'.**
- 4. Show employees who earn more than the average salary of their department.**
- 5. Display employees working on projects with the highest budget.**
- 6. Show departments that have employees earning more than ₹60,000.**